

Аномалии в потоковых данных

алгоритмы под капотом

Обо мне

- Старший специалист по машинному обучению
- deep learning engineer
- NLP, CV, anomaly detection
- Open source contributor
- Амбассадор Яндекс Практикума



Аномалии

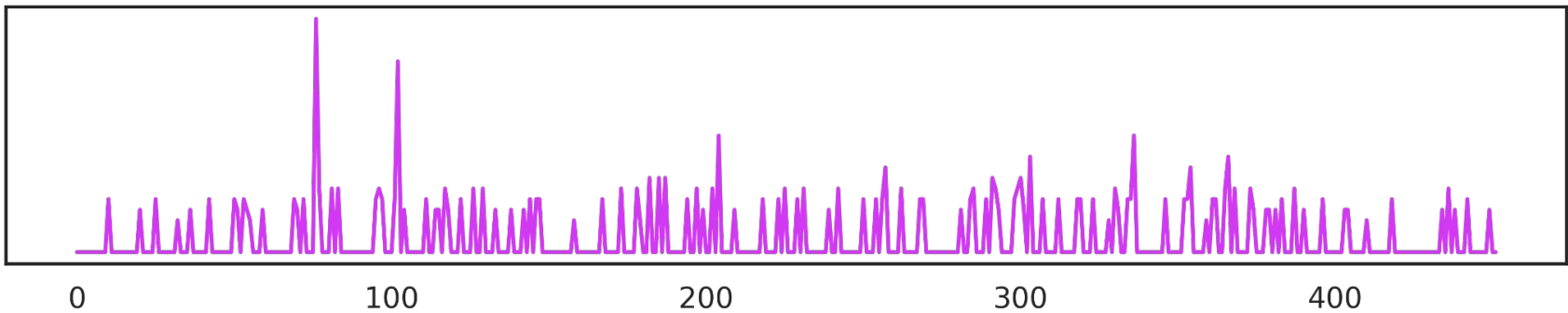
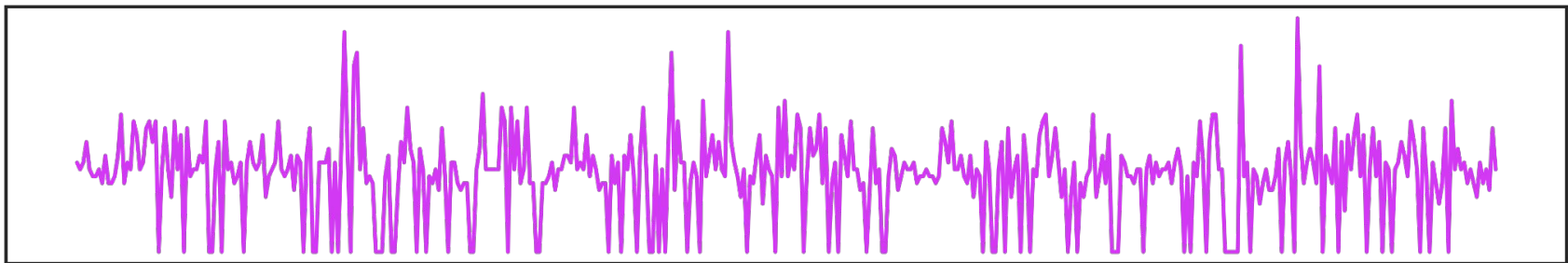
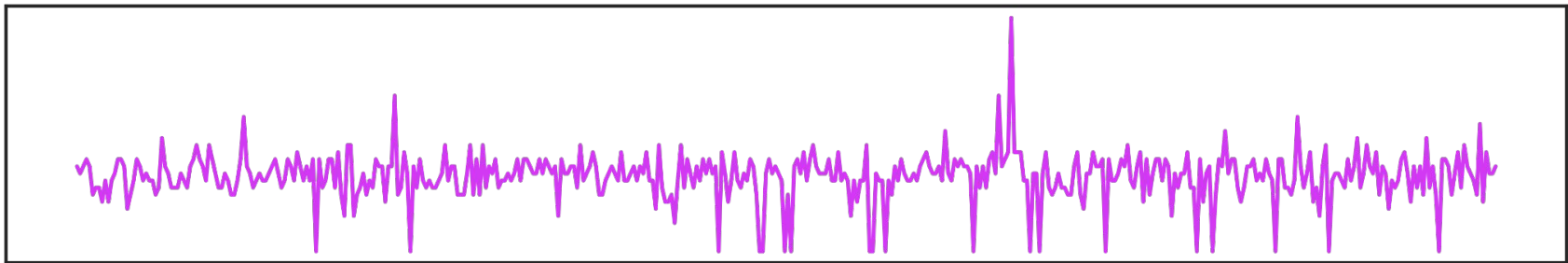


SCAMMER

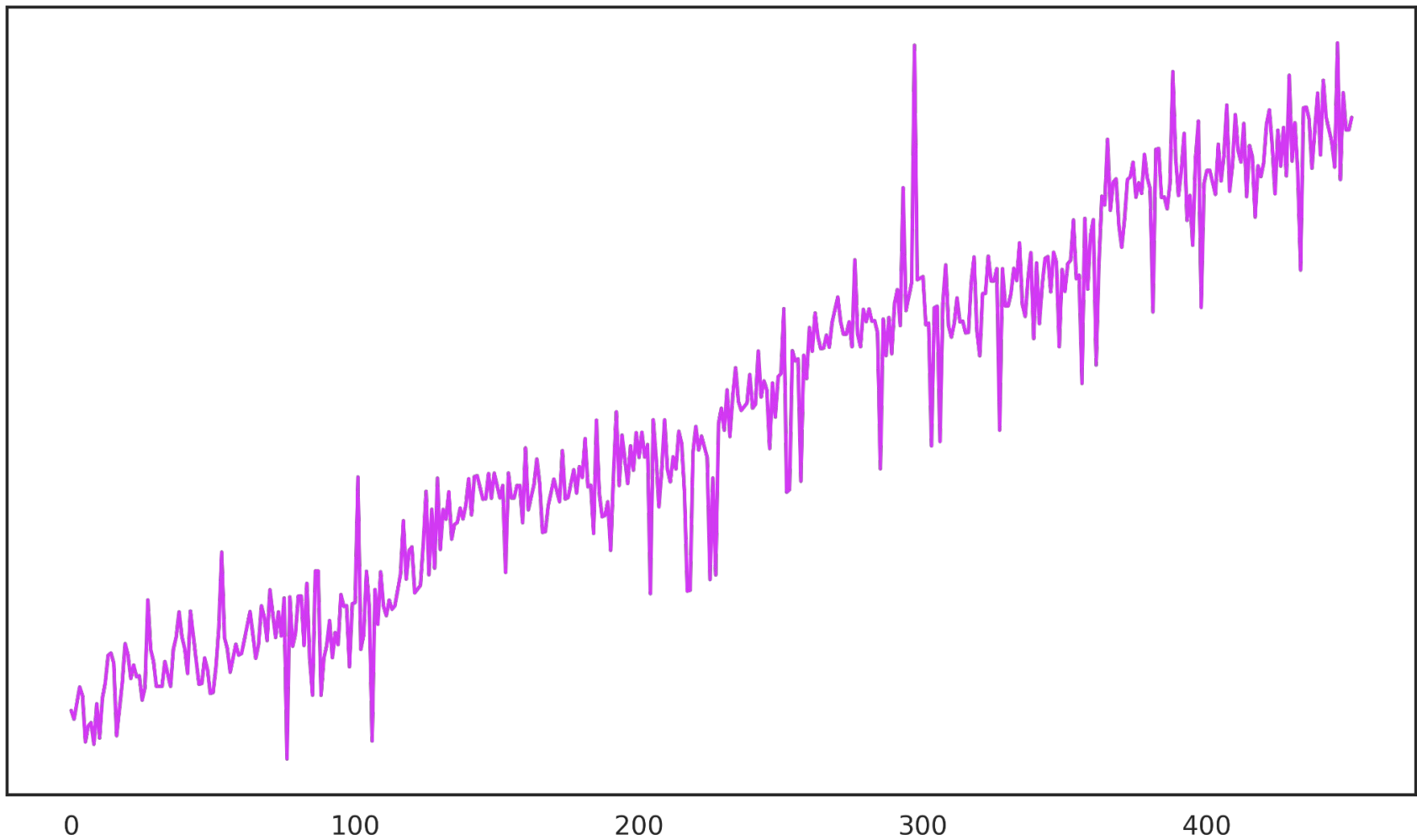
попугаи	удаваы
3.248357	-2.874754
2.930868	-2.826776
3.323844	-3.340012
3.761515	-2.883873
2.882923	-2.853464
2.882932	-3.357176
3.789606	-2.067113
...	...

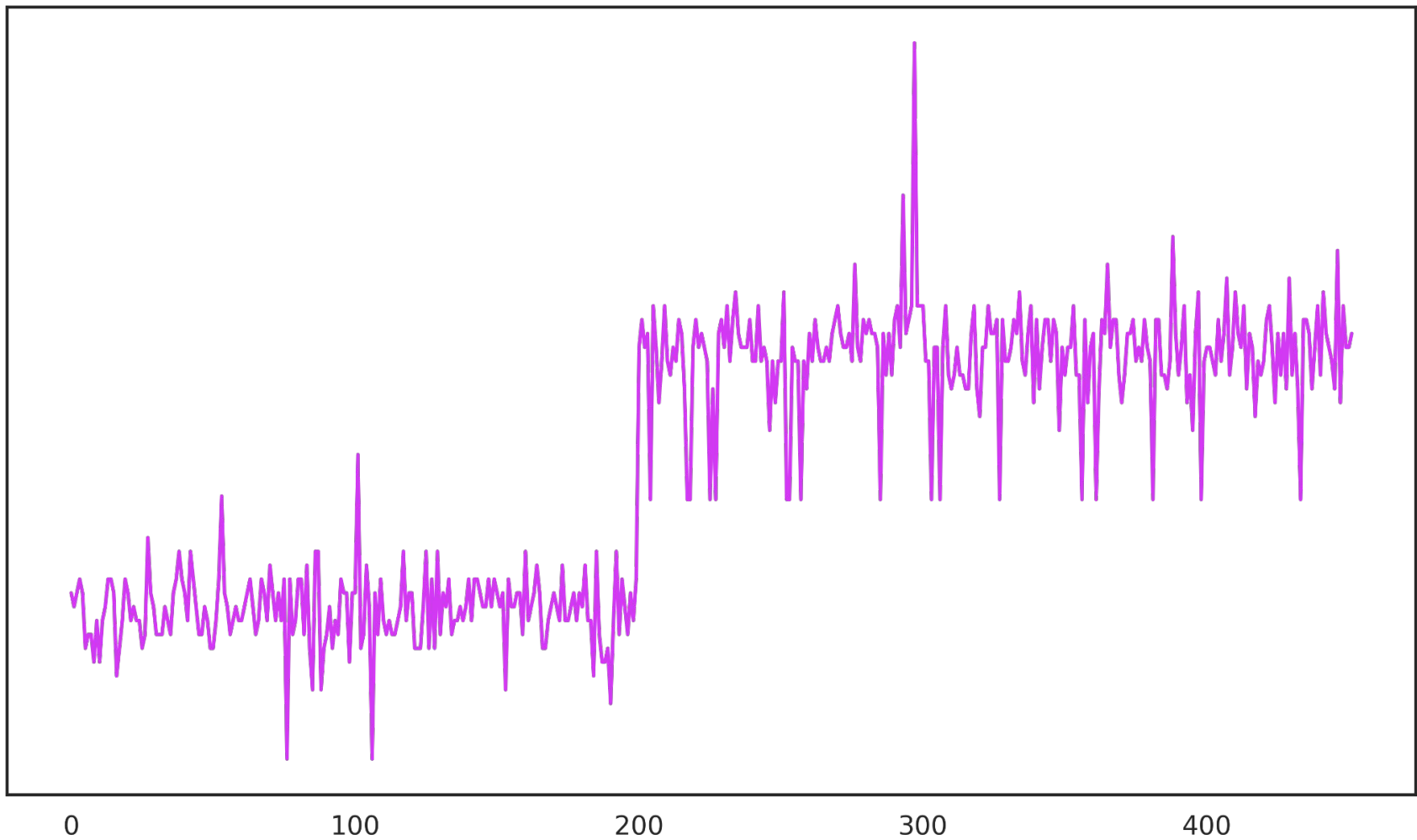
попугаи	удаваы
3.248357	-2.874754
2.930868	-2.826776
3.323844	-3.340012
3.761515	-2.883873
2.882923	-2.853464
2.882932	-3.357176
3.789606	-2.067113
...	...

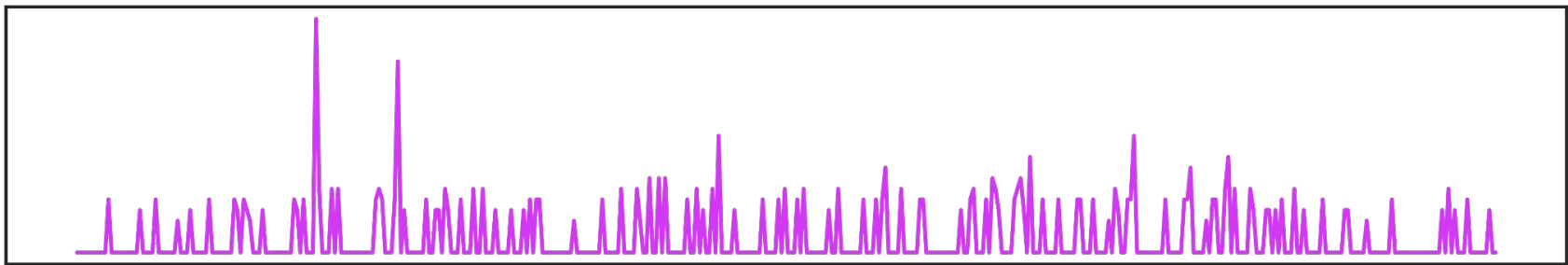
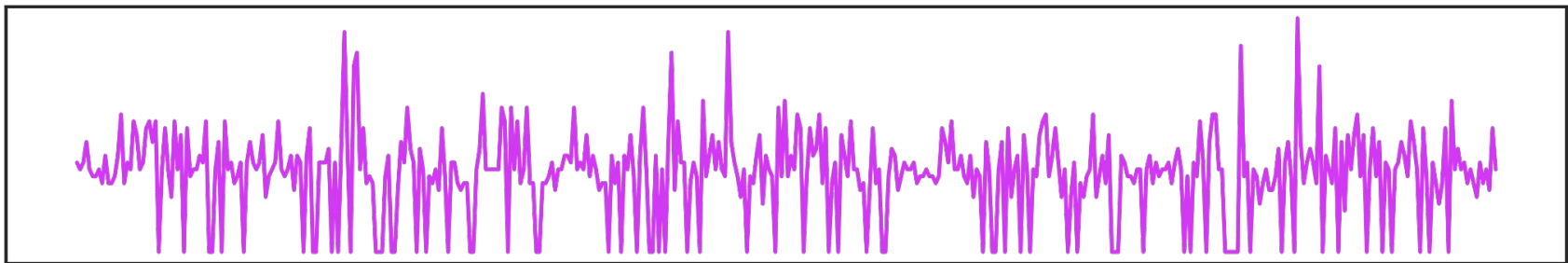
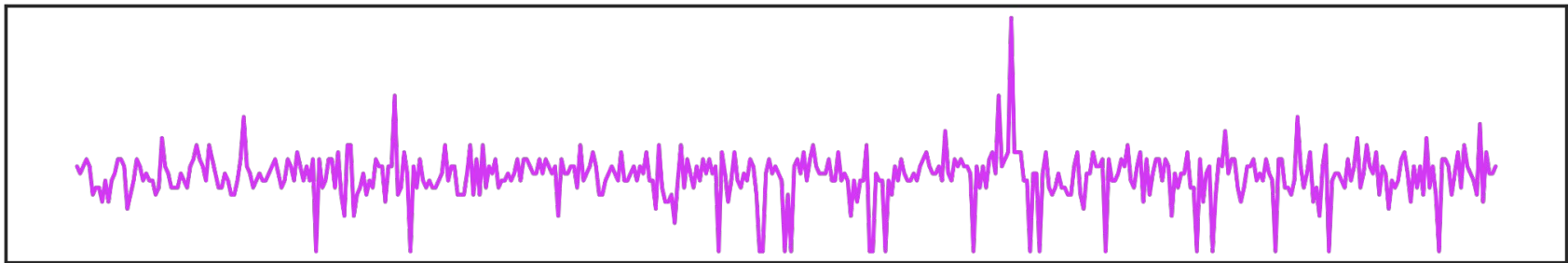
индекс аномальности
-0.119150
-0.131275
-0.108670
-0.065472
-0.128972
-0.120056
-0.012170
...



Threshold







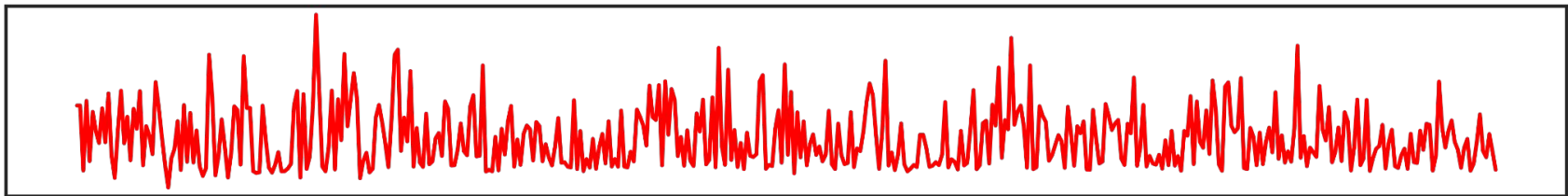
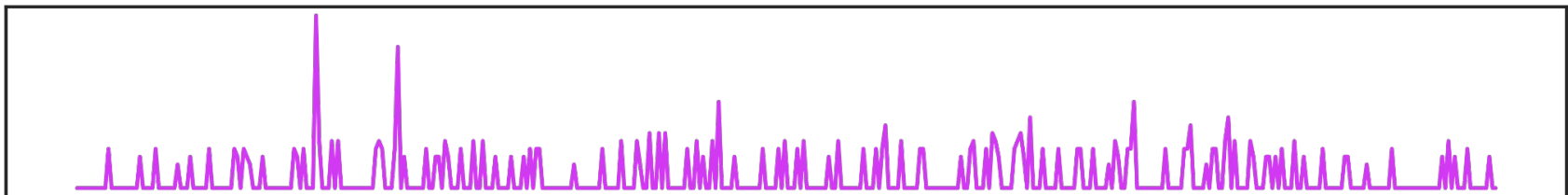
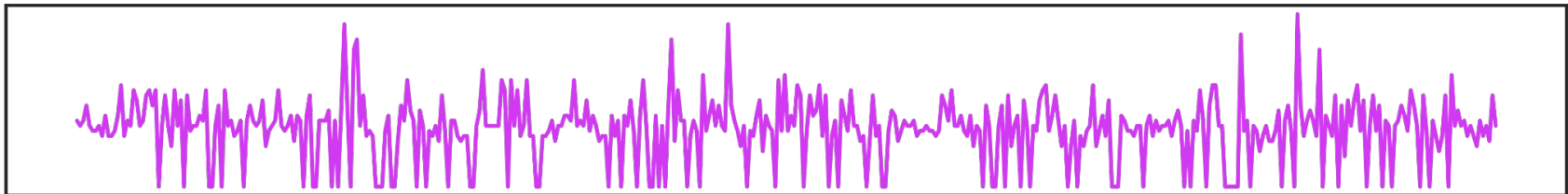
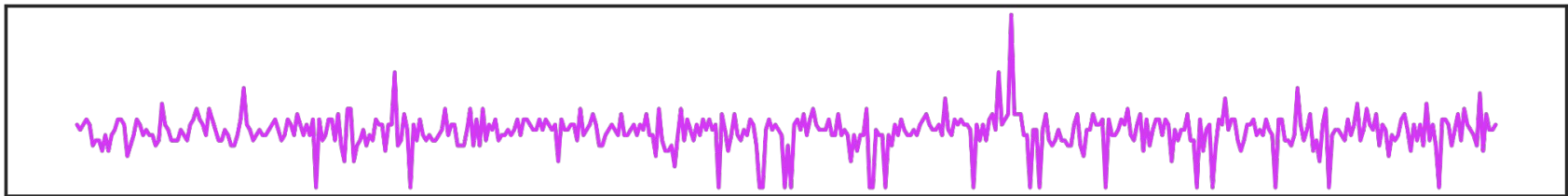
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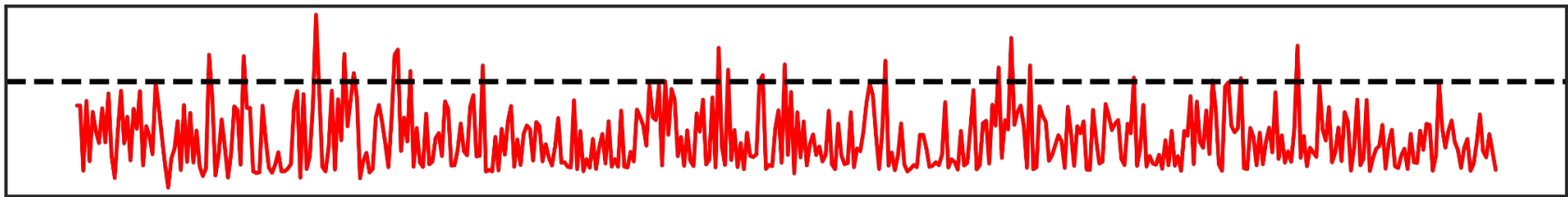
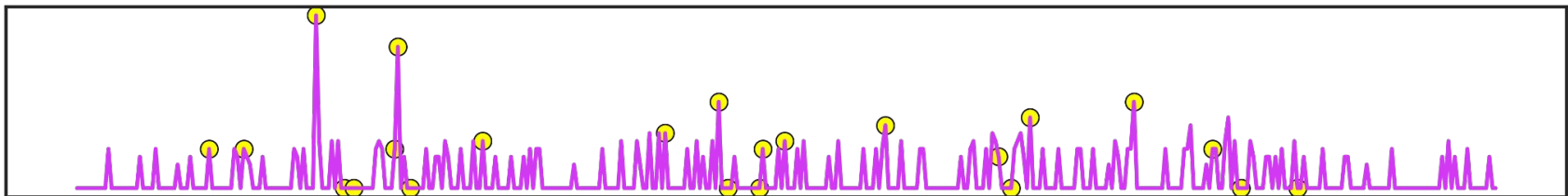
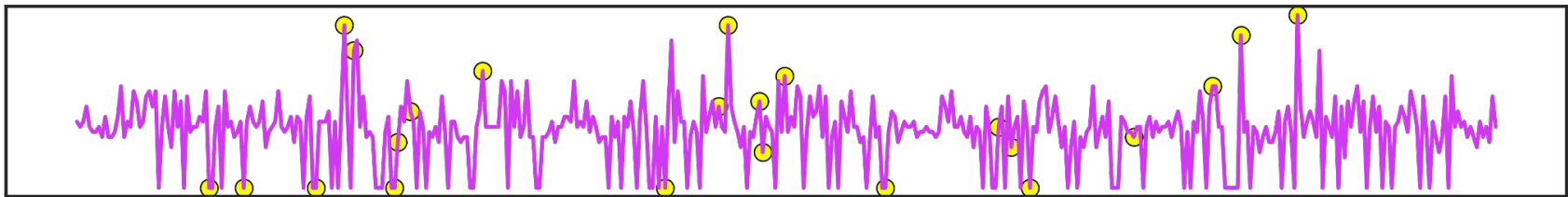
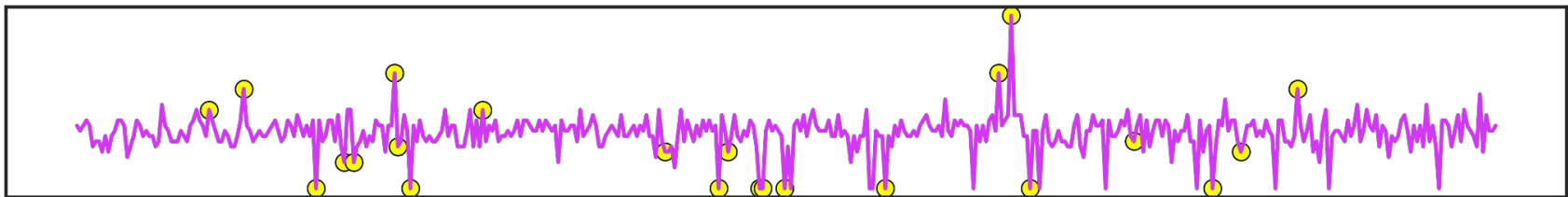
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Потоковые данные

```
pip install pysad
```

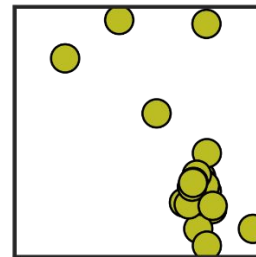
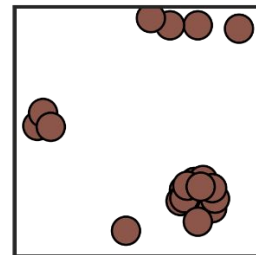
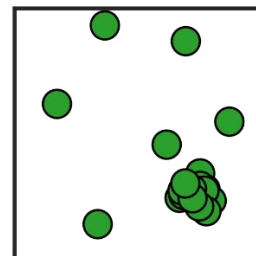
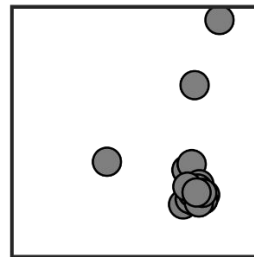
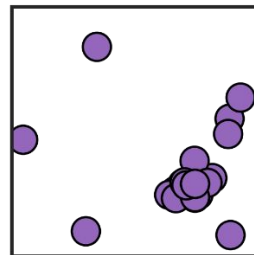
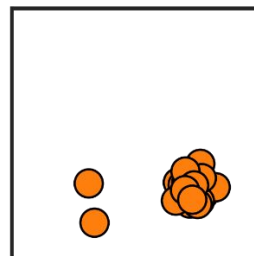
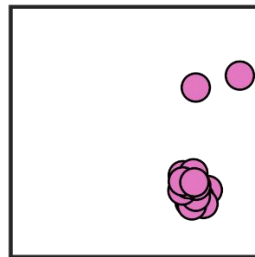
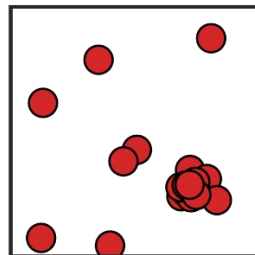
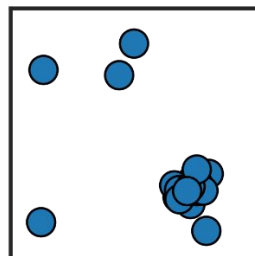
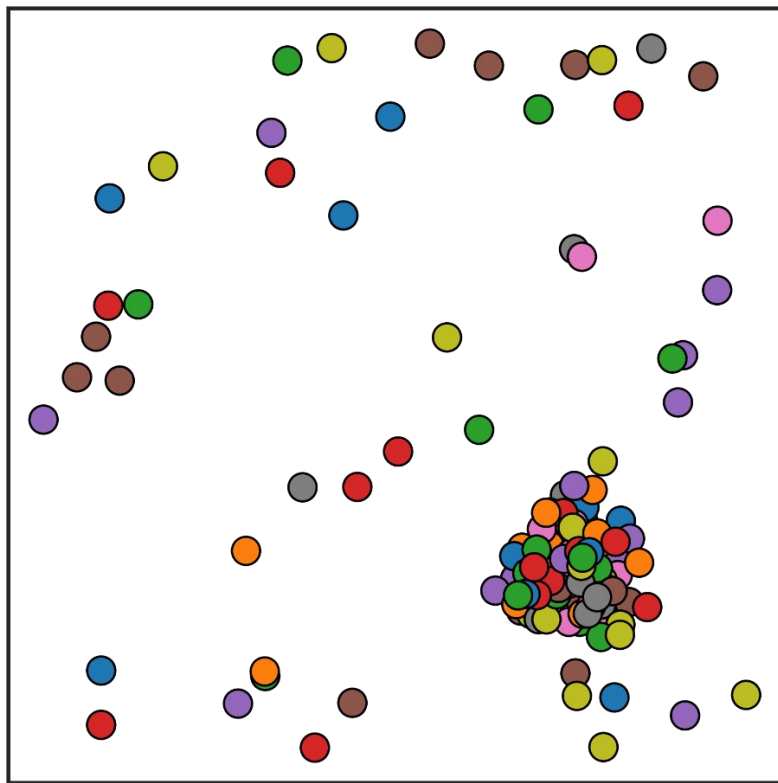
```
from pysad.models import IForestASD
```

```
model = IForestASD()
```

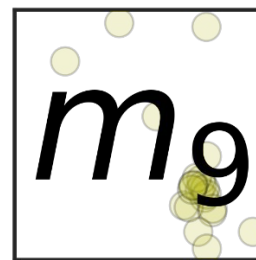
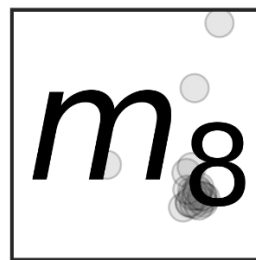
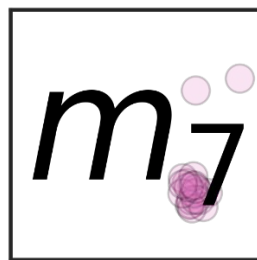
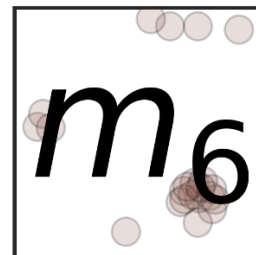
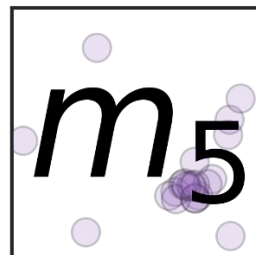
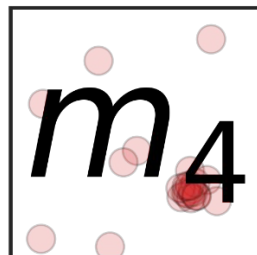
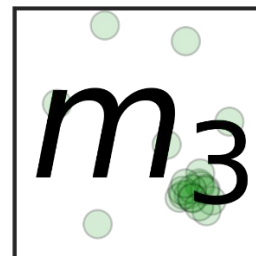
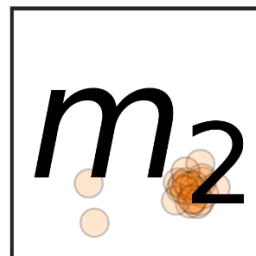
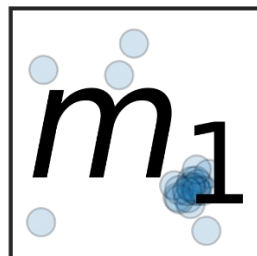
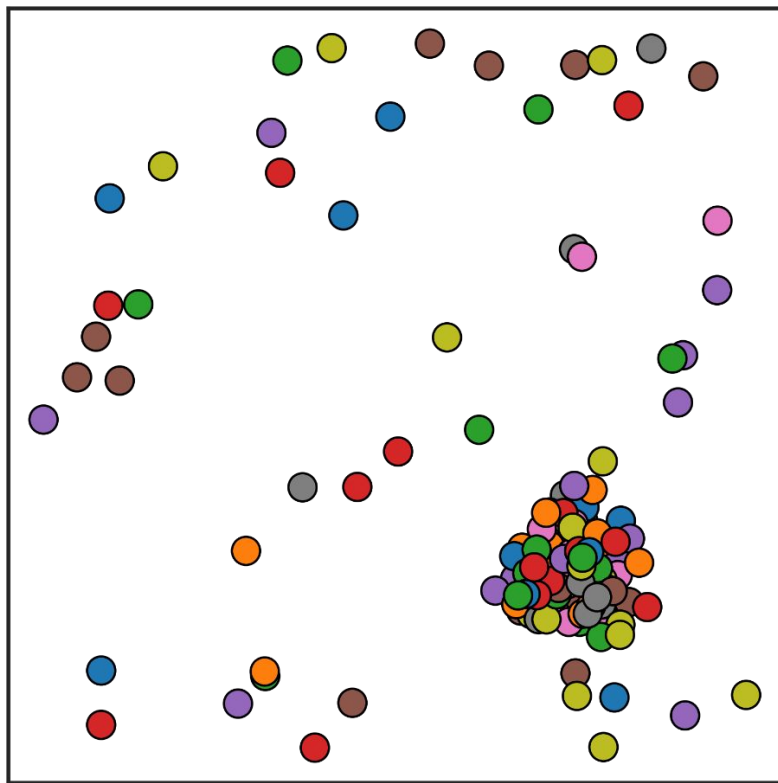
```
for x in streaming_data:  
    anomaly_score = model.fit_score_partial(x)
```

Ансамбли

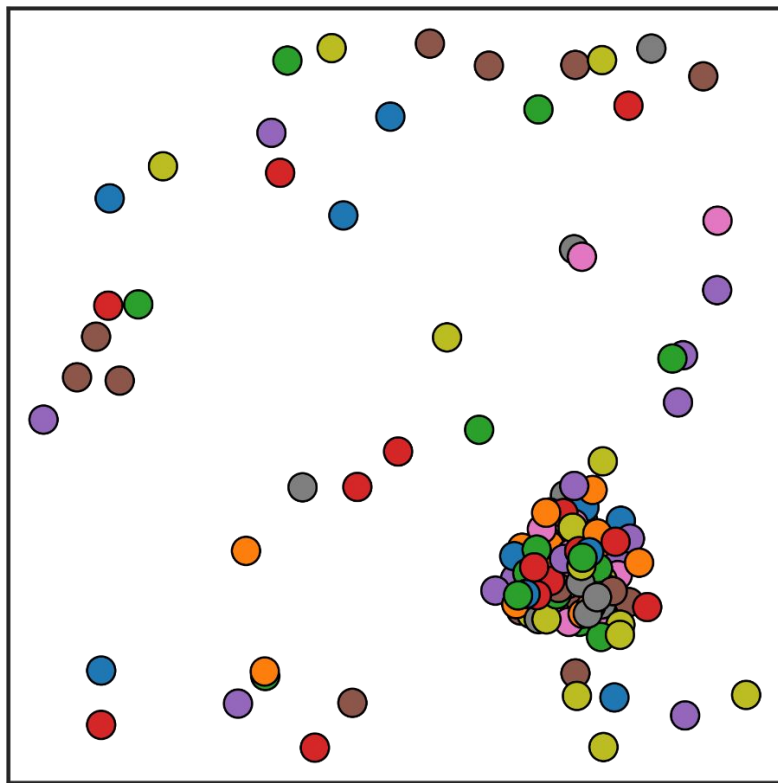
Делим

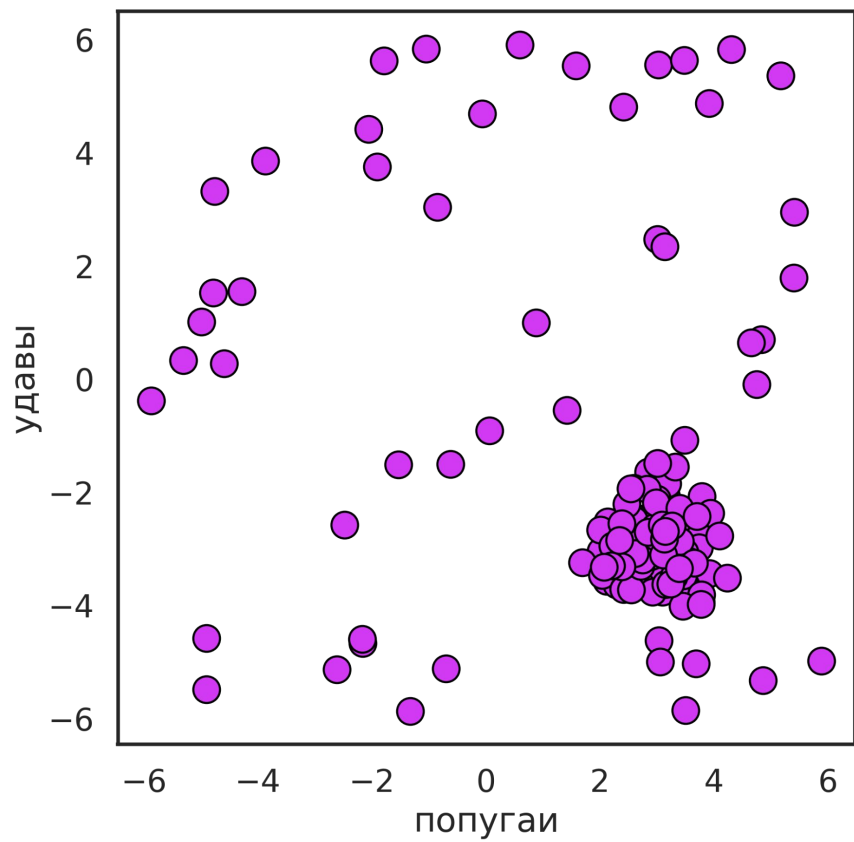


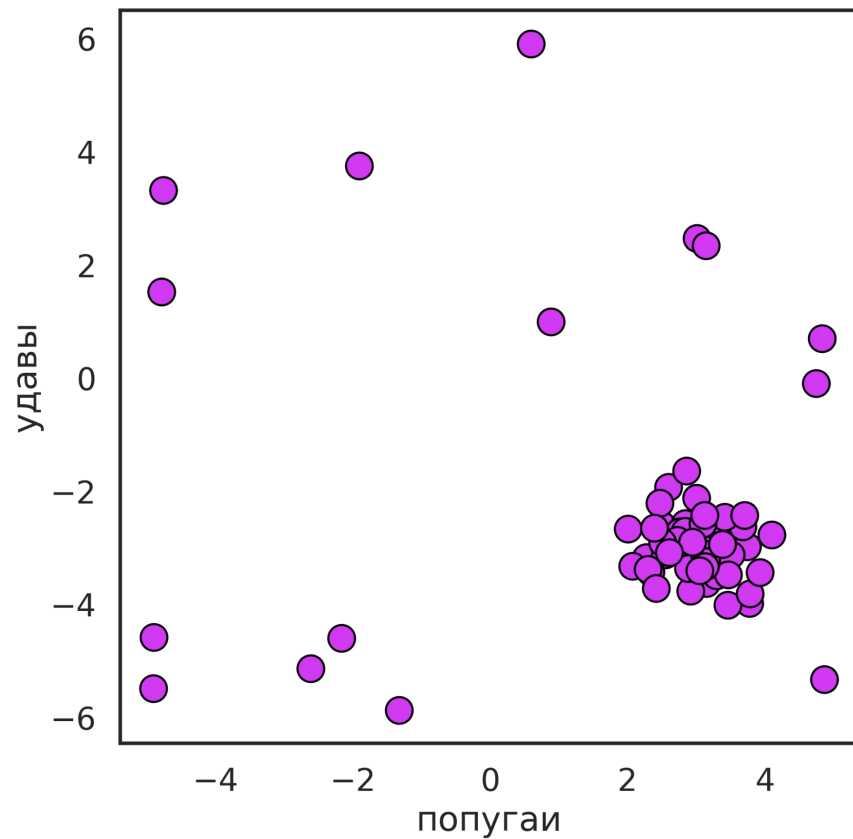
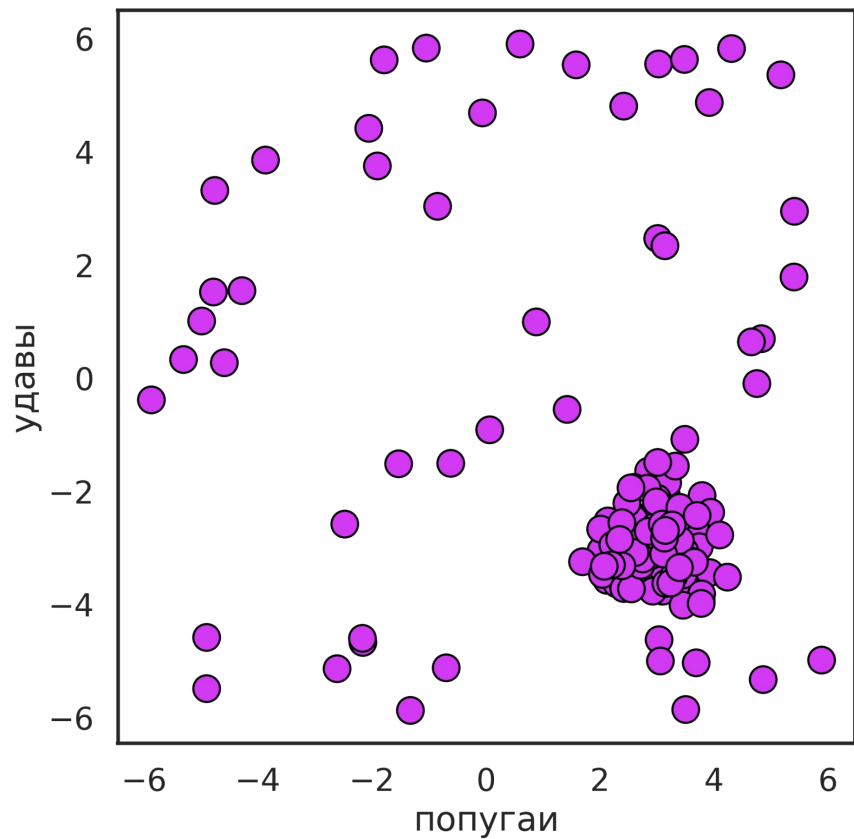
Обучаем



Усредняем







```
In [1]: import random
```

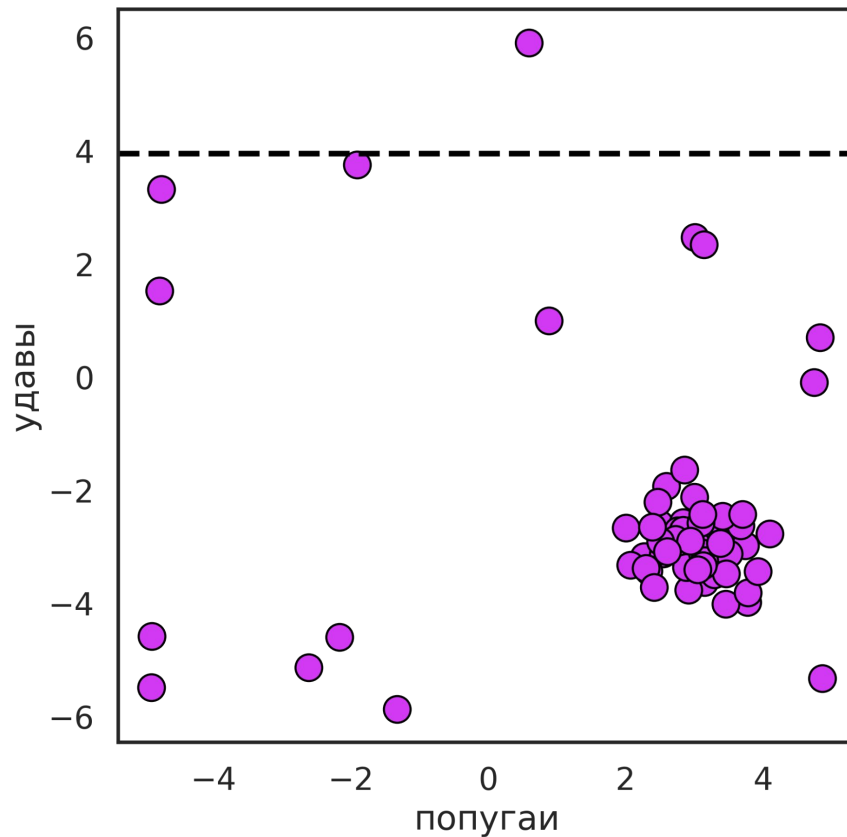
```
In [2]: axes = ['попугаи', 'удавы']
```

```
In [3]: random.choice(axes)
```

```
Out[3]: 'удавы'
```

```
In [4]: random.uniform(df['удавы'].min(),  
df['удавы'].max())
```

```
Out[4]: 3.954197818641566
```



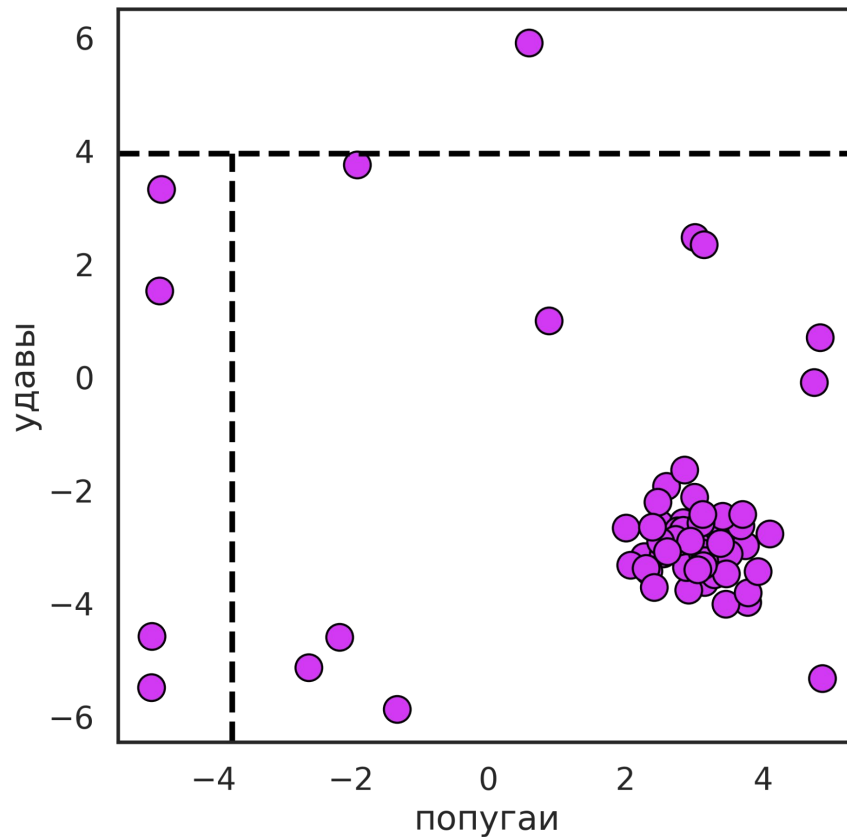
...

```
In [5]: random.choice(axes)
```

```
Out[5]: 'попугаи'
```

```
In [6]: random.uniform(df_b['попугаи'].min(),  
df_b['попугаи'].max())
```

```
Out[6]: -3.7345546743319455
```



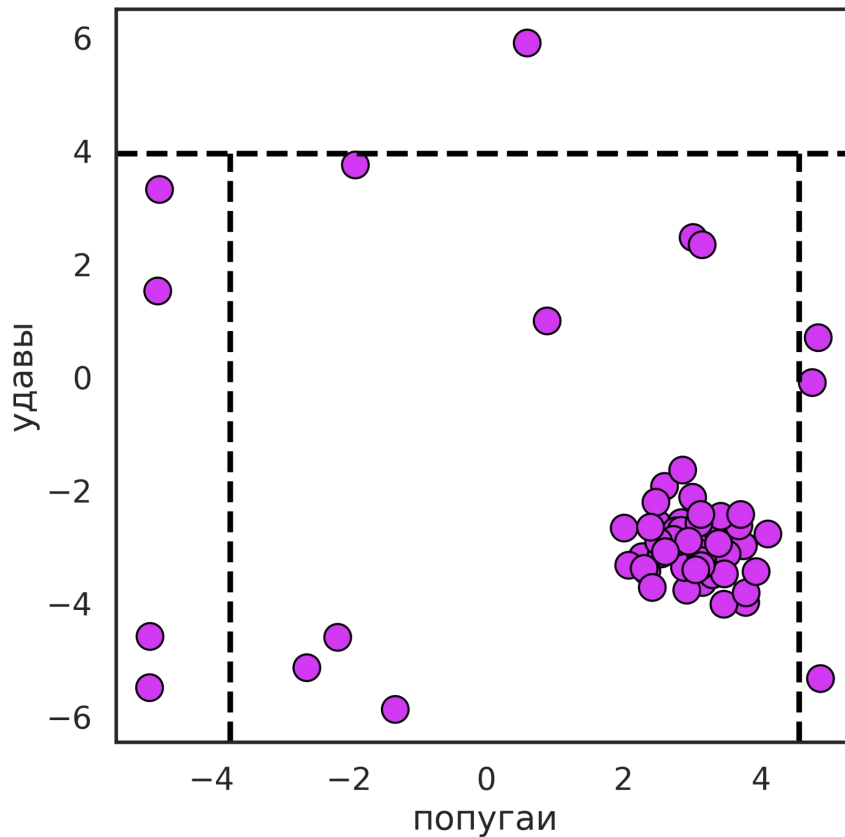
...

```
In [7]: random.choice(axes)
```

```
Out[7]: 'попугаи'
```

```
In [8]: random.uniform(df_bl['попугаи'].min(),  
df_bl['попугаи'].max())
```

```
Out[8]: 4.55352143693694
```



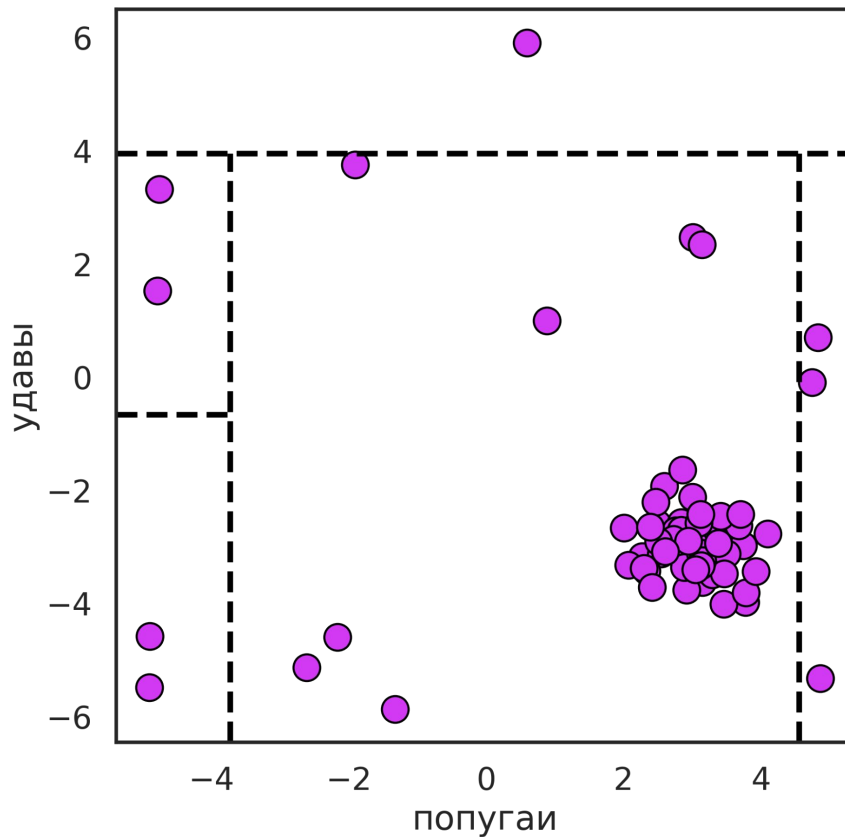
...

```
In [9]: random.choice(axes)
```

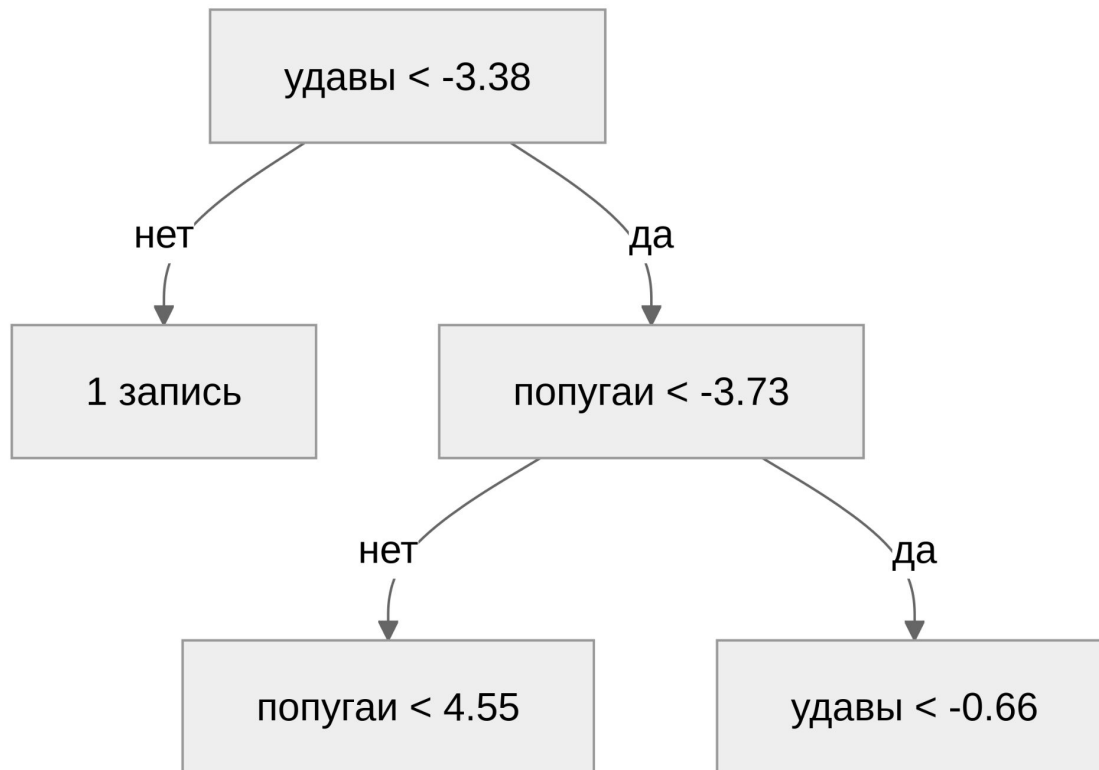
```
Out[9]: 'удавы'
```

```
In [10]: random.uniform(df_br['удавы'].min(),  
df_br['удавы'].max())
```

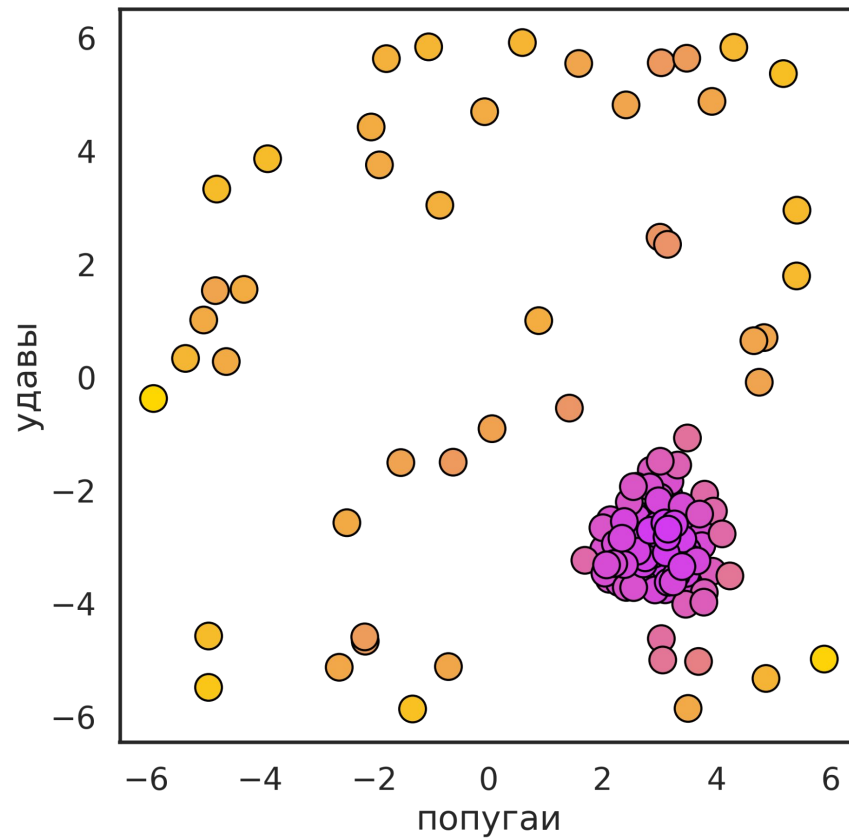
```
Out[10]: -0.6572094533790986
```



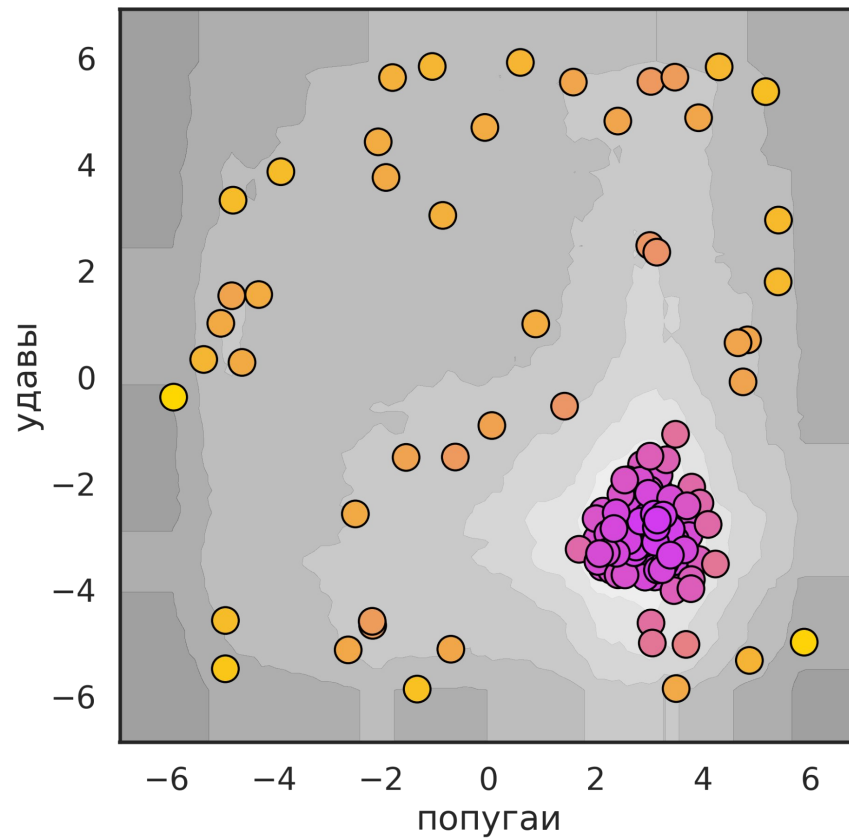
iTree



Isolation Forest



Isolation Forest

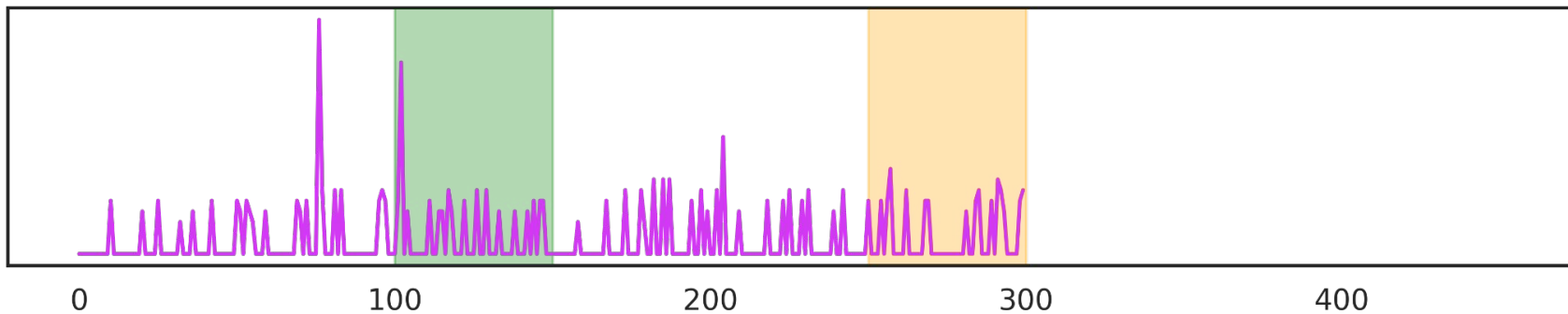
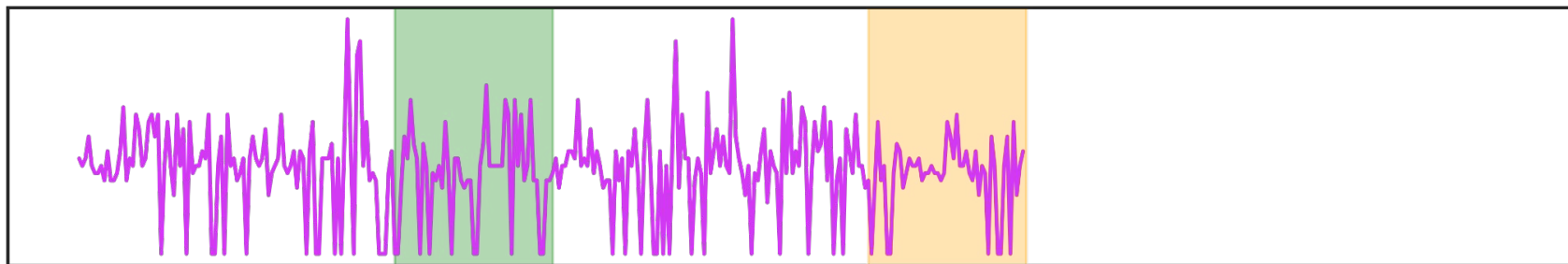
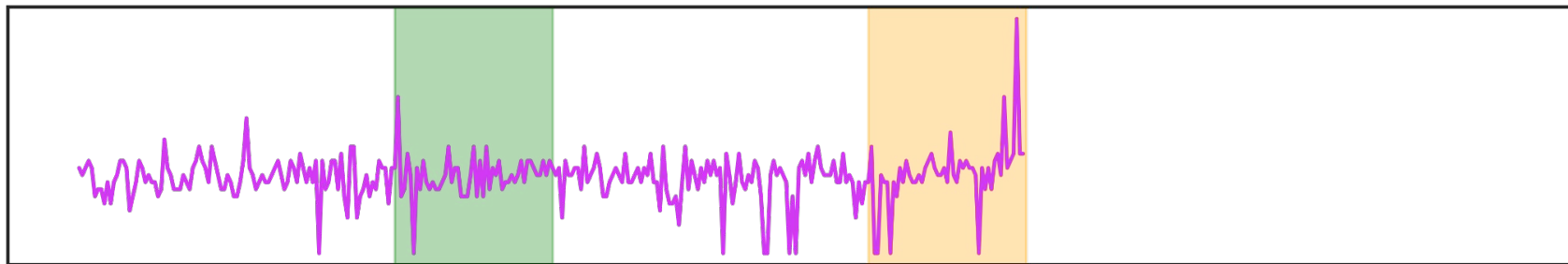


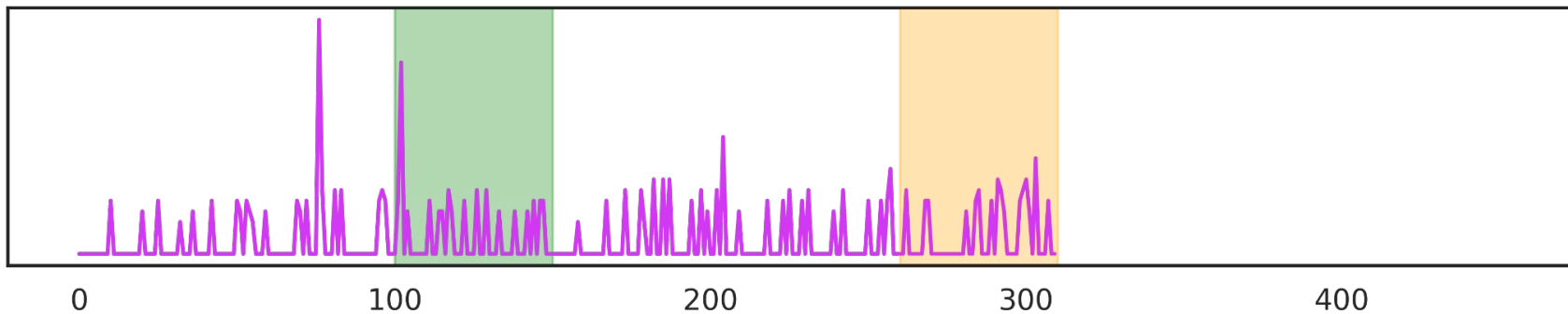
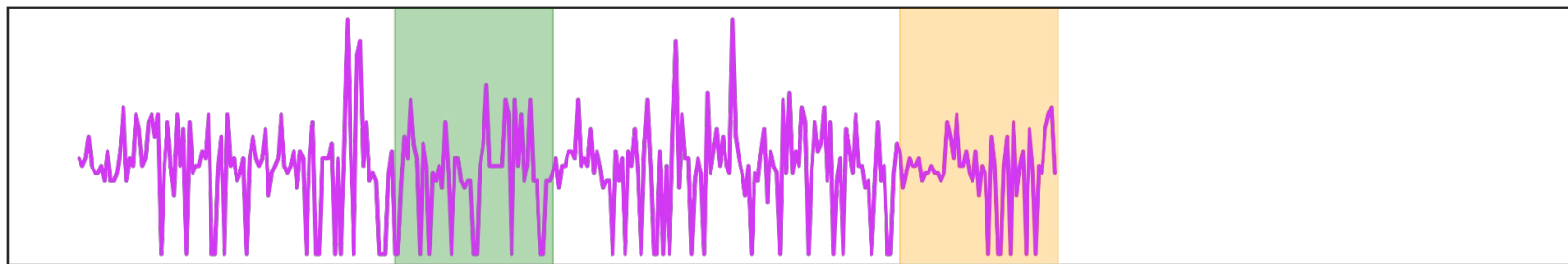
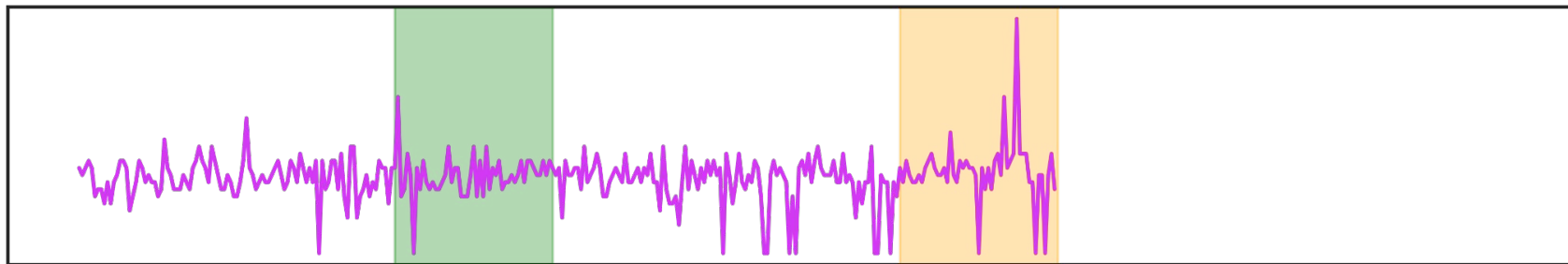
Алгоритм

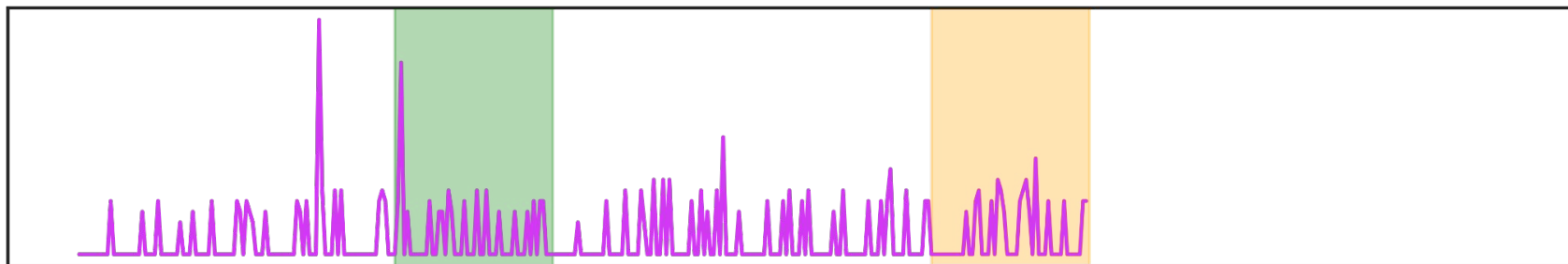
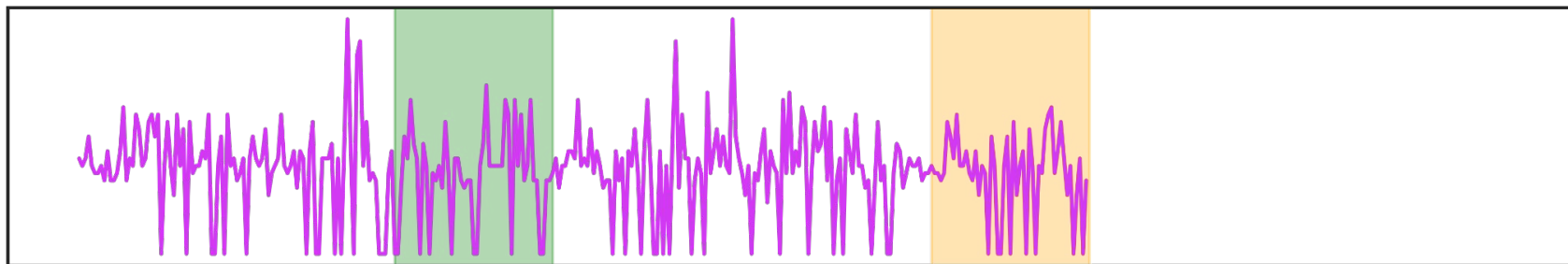
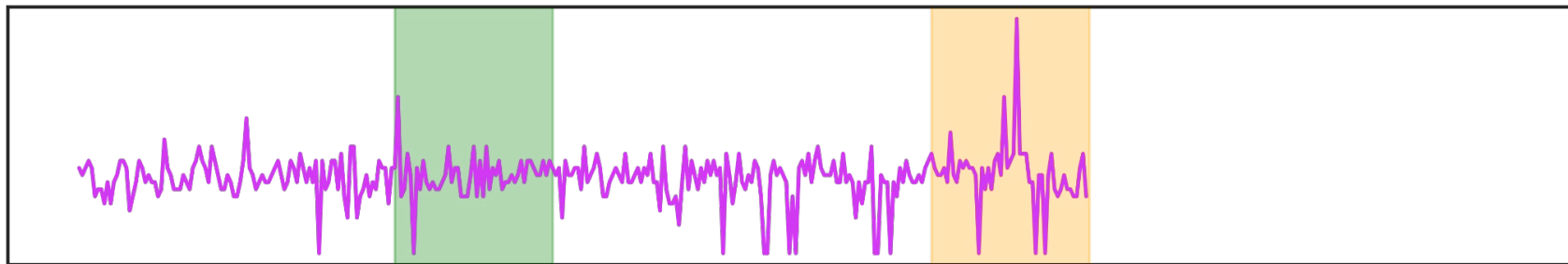
Fei Tony Liu, Kai Ming Ting, and Zhi-Hua Zhou. Isolation forest. In *Data Mining, 2008. ICDM'08. Eighth IEEE International Conference on*, 413–422. IEEE, 2008.

Потоковые данные

Zhiguo Ding and Minrui Fei. An anomaly detection approach based on isolation forest algorithm for streaming data using sliding window. *IFAC Proceedings Volumes*, 46(20):12–17, 2013.







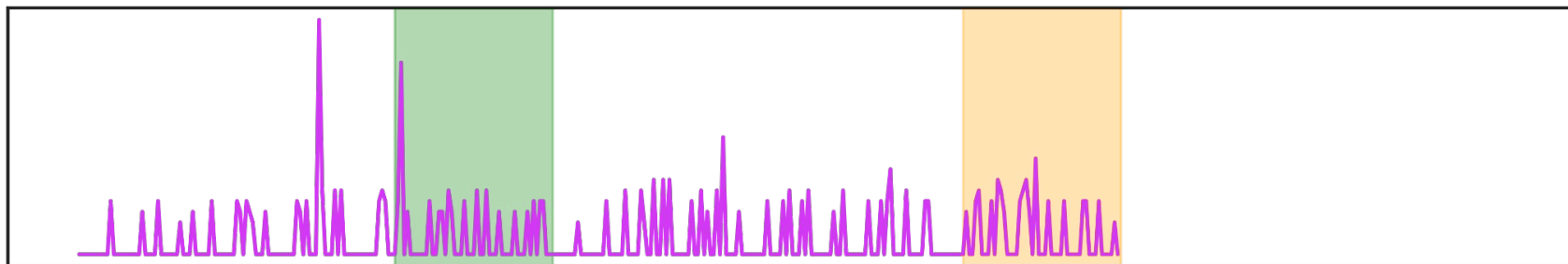
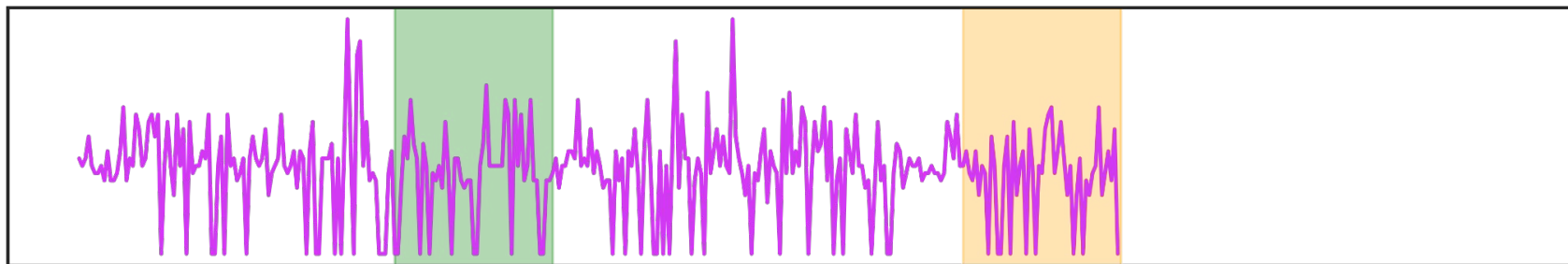
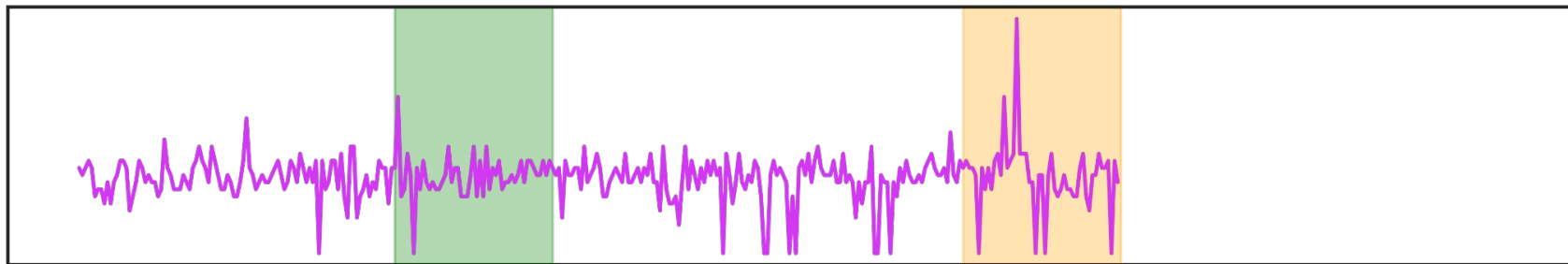
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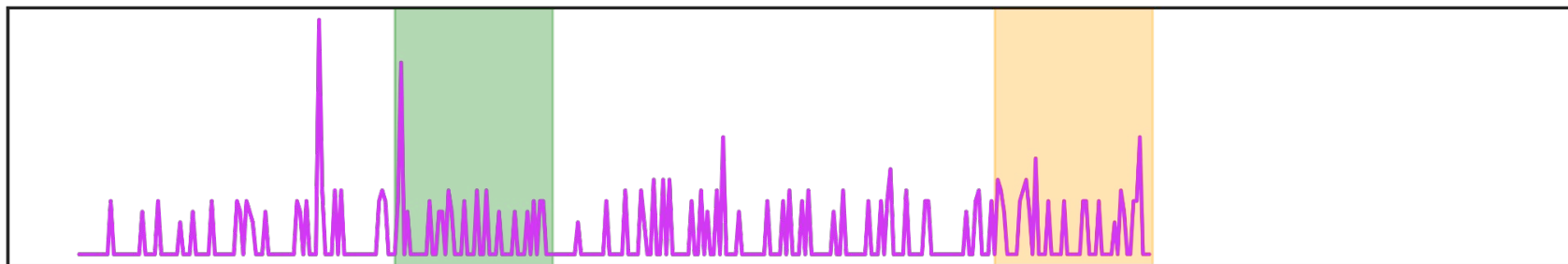
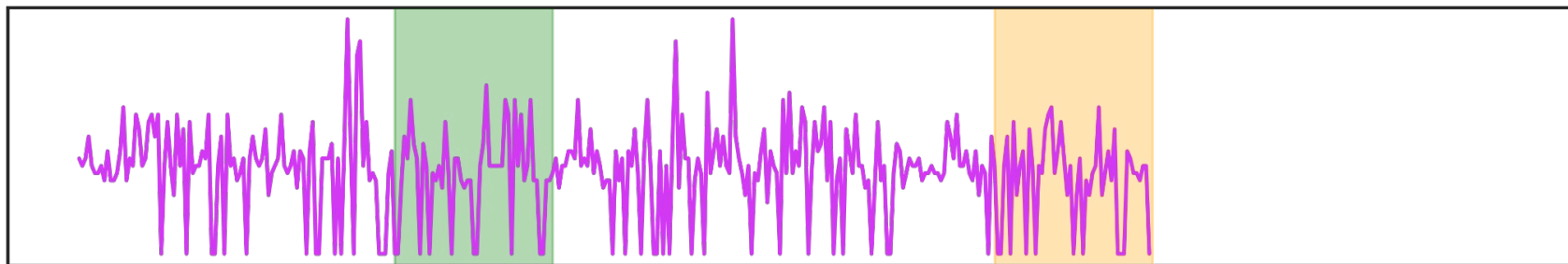
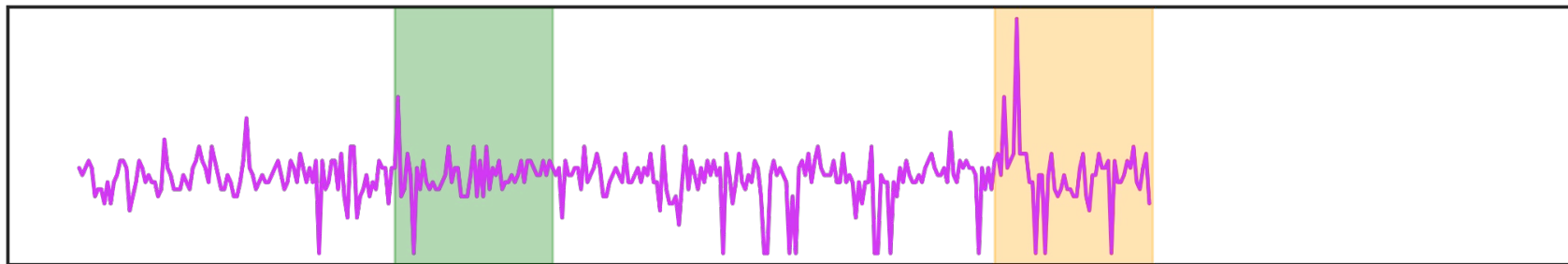
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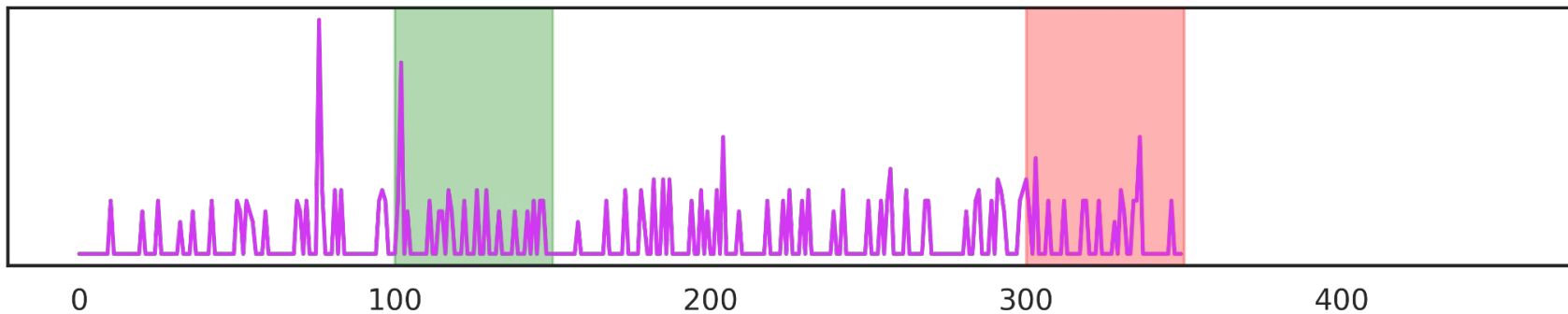
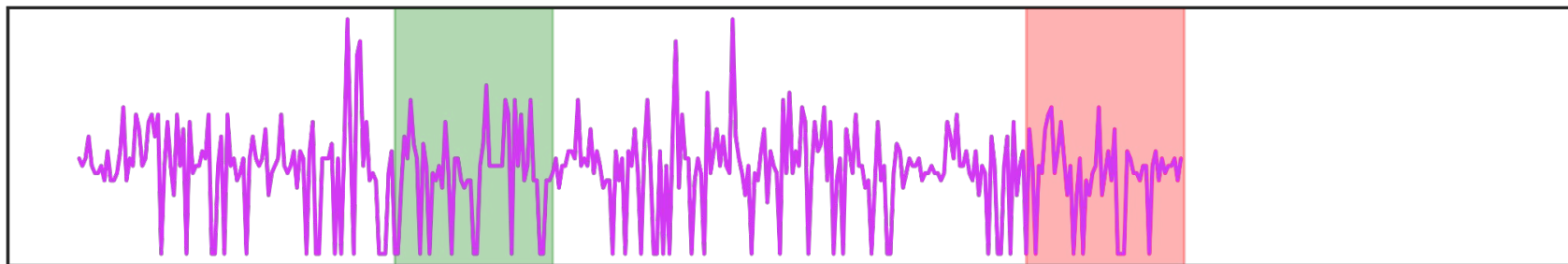
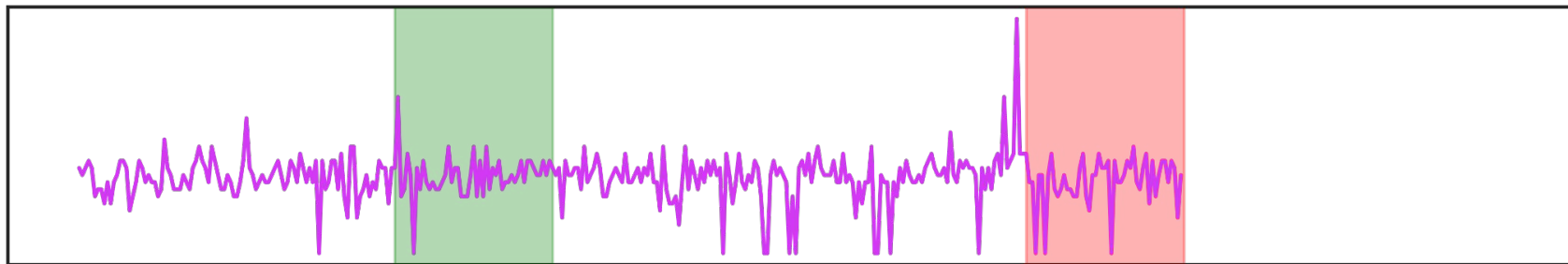
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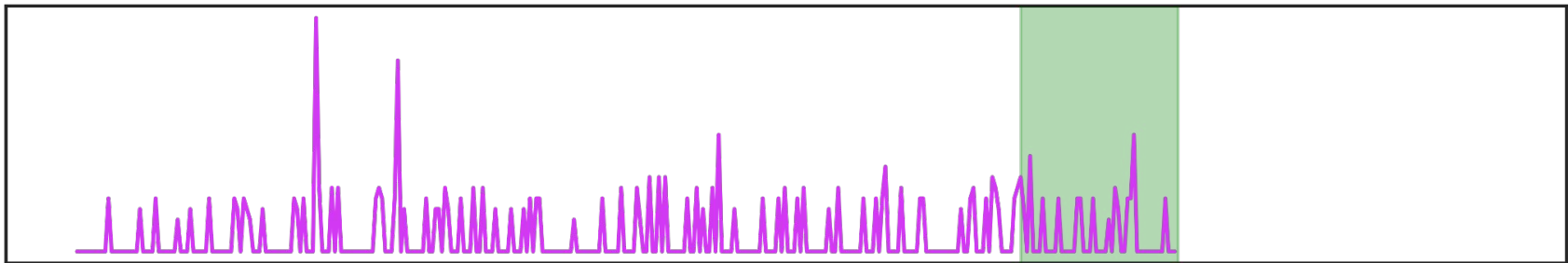
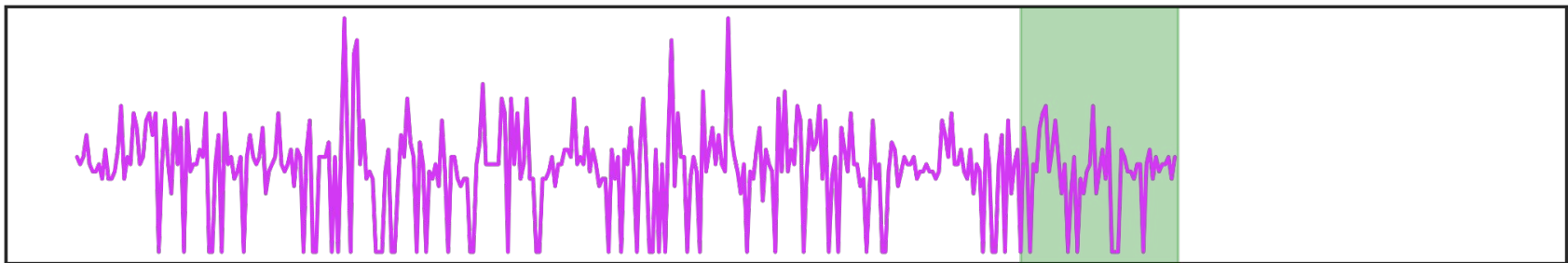
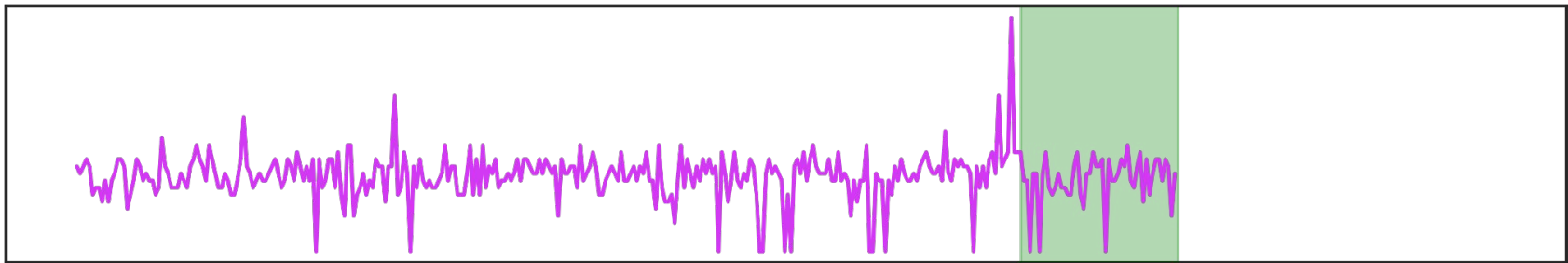
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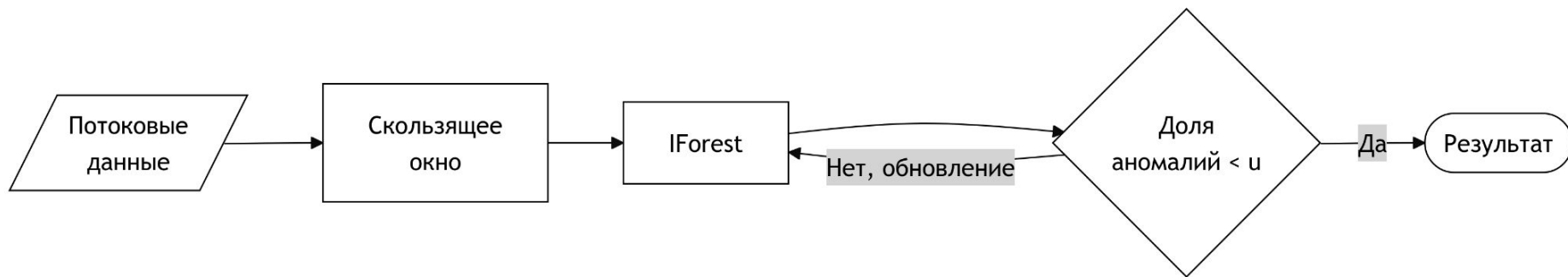
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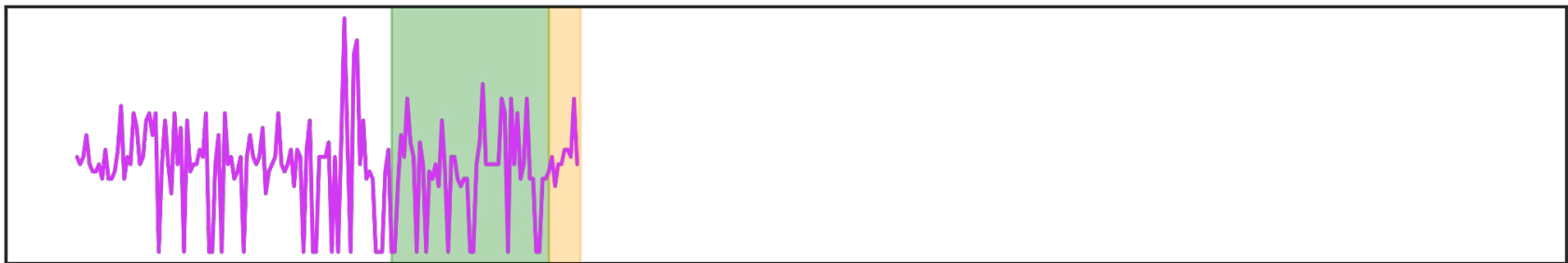
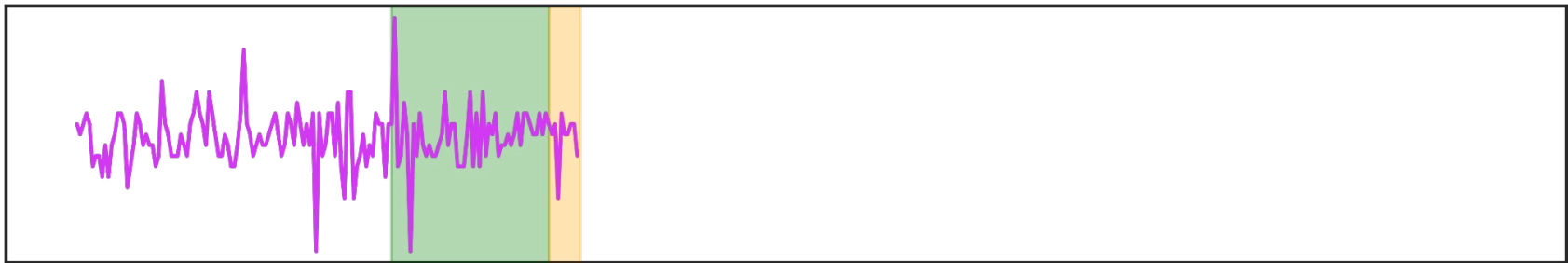
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Алгоритм





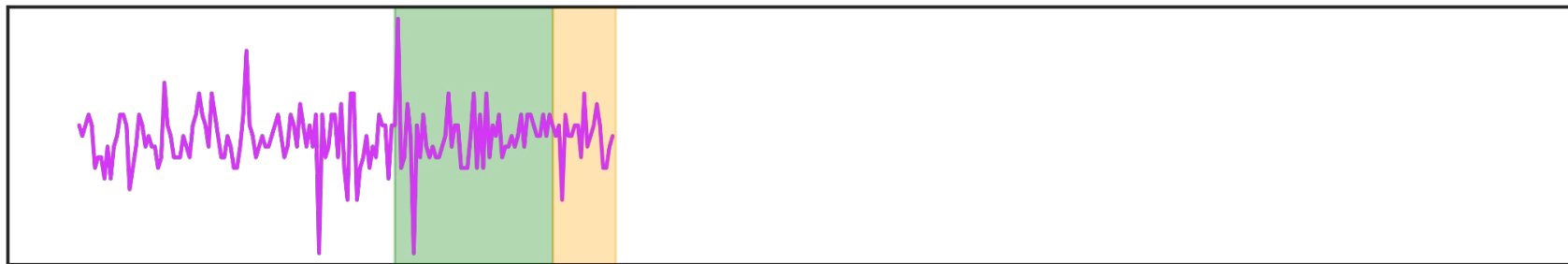
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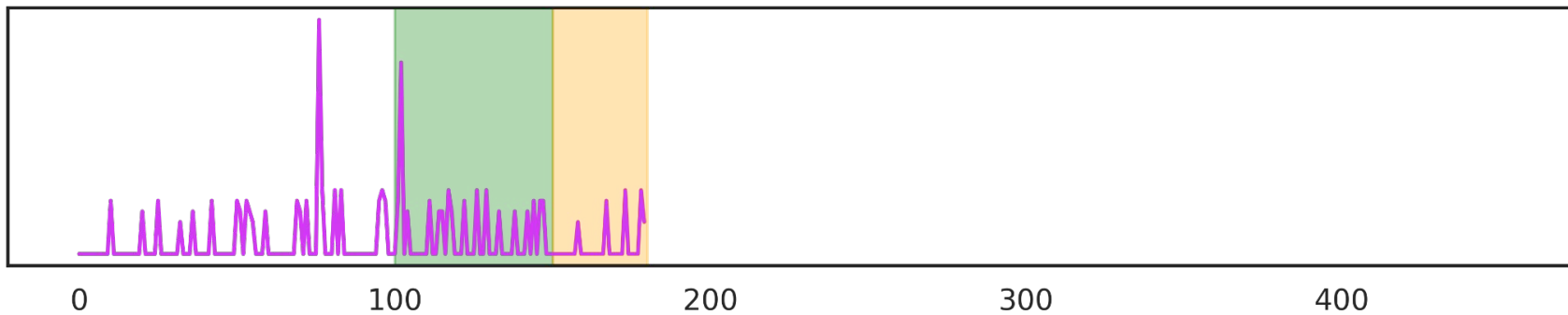
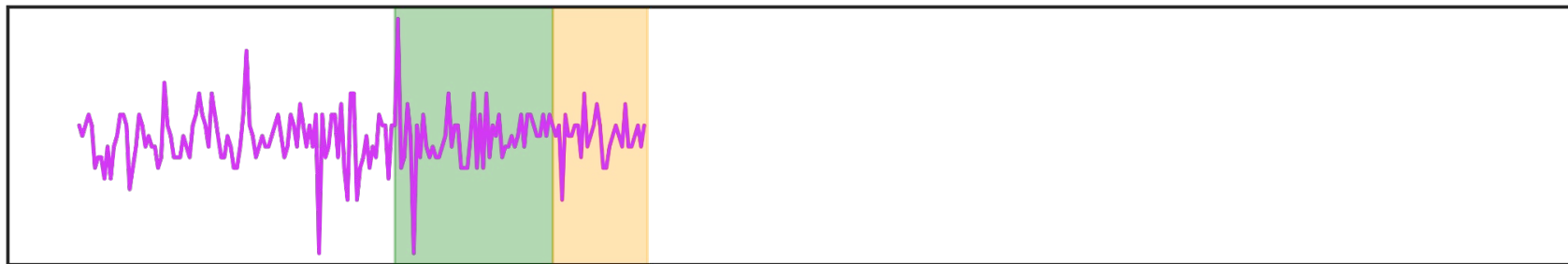
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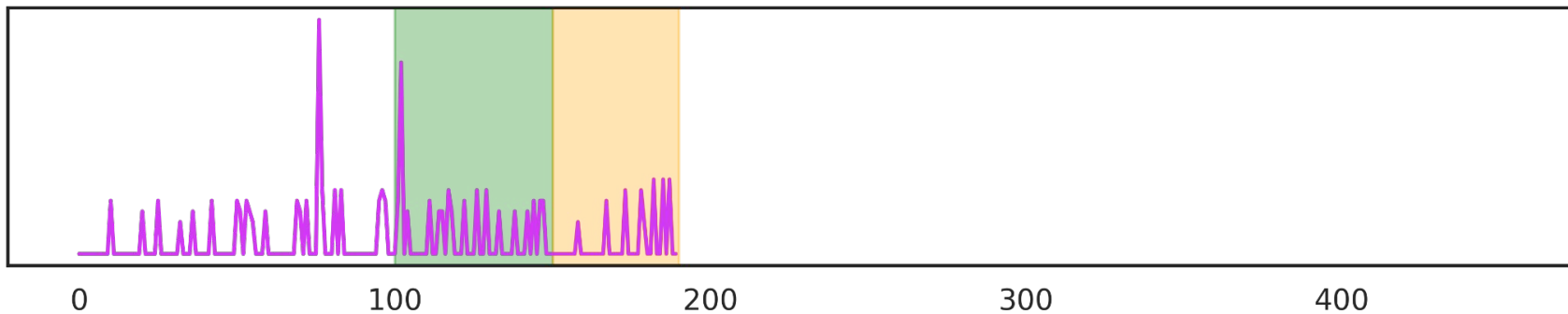
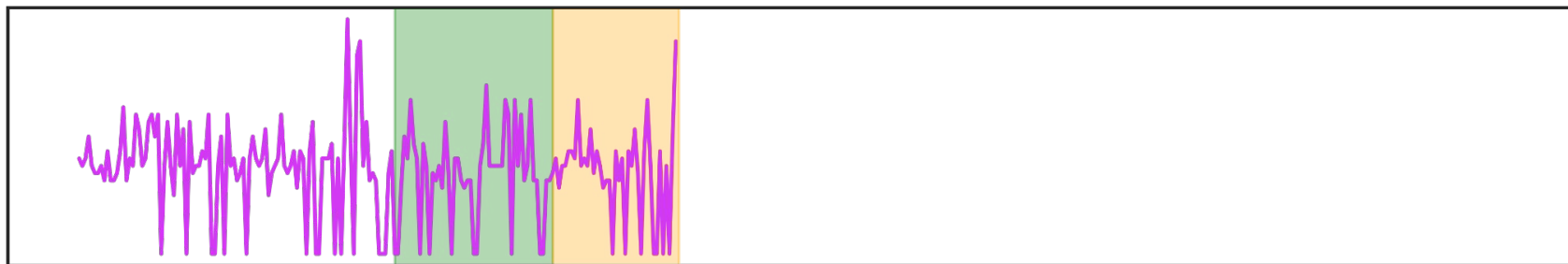
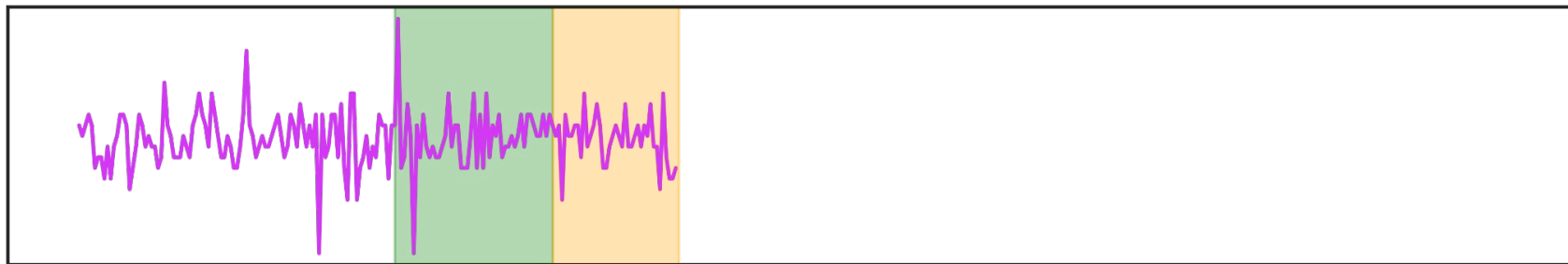
200

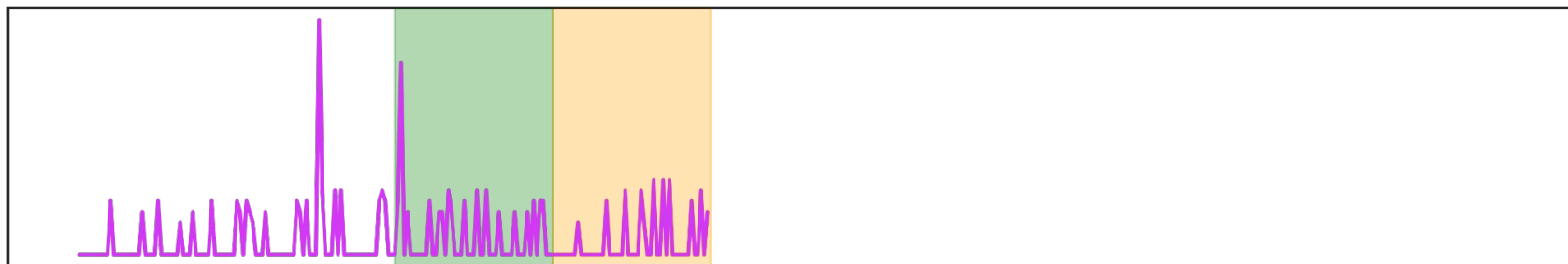
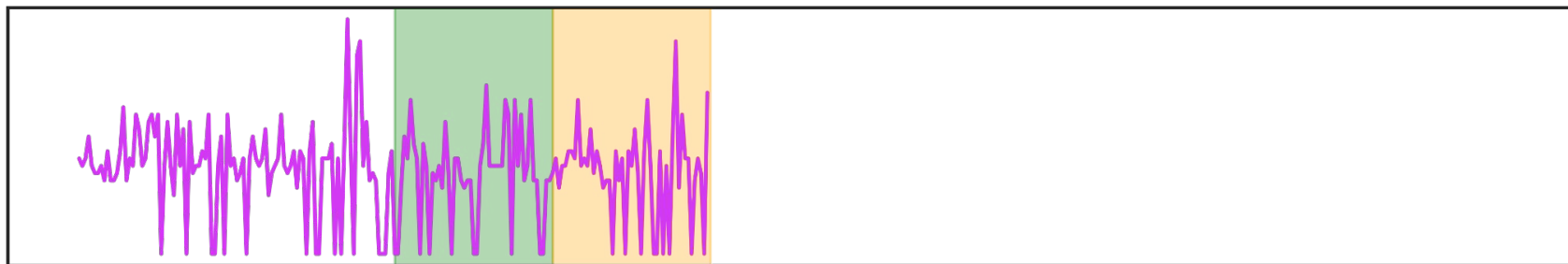
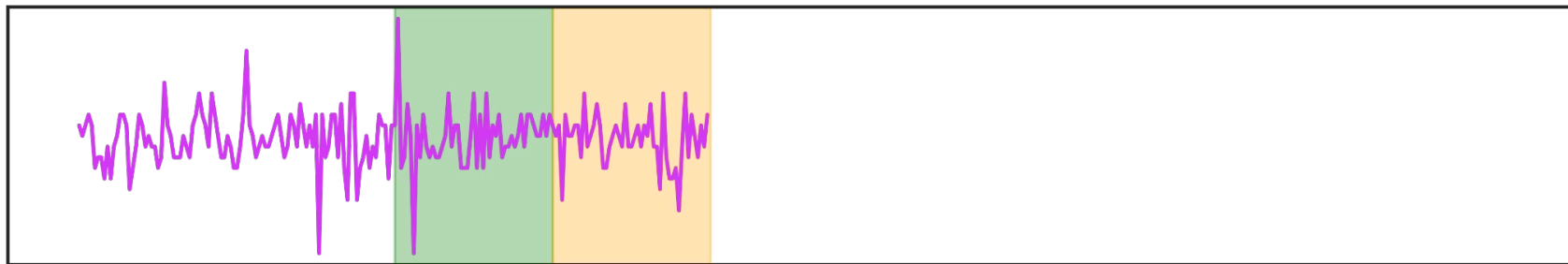
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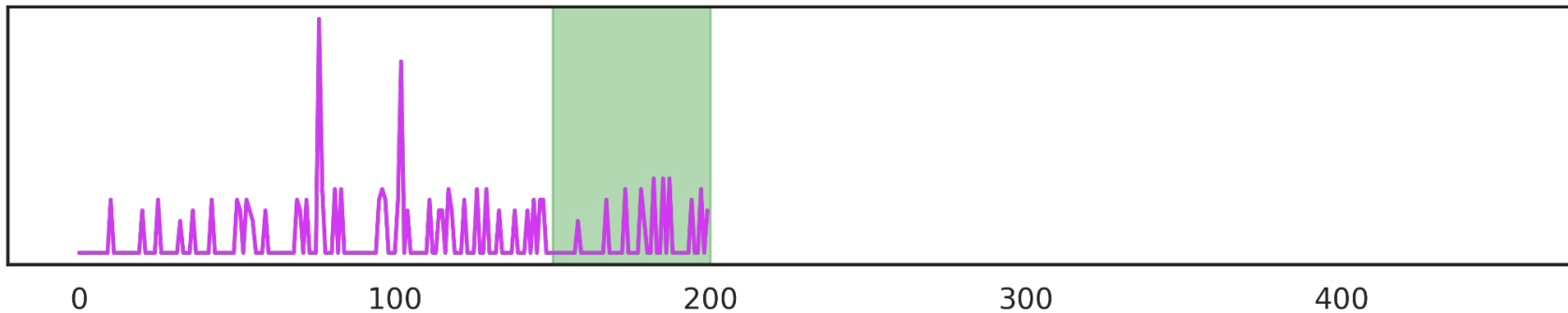
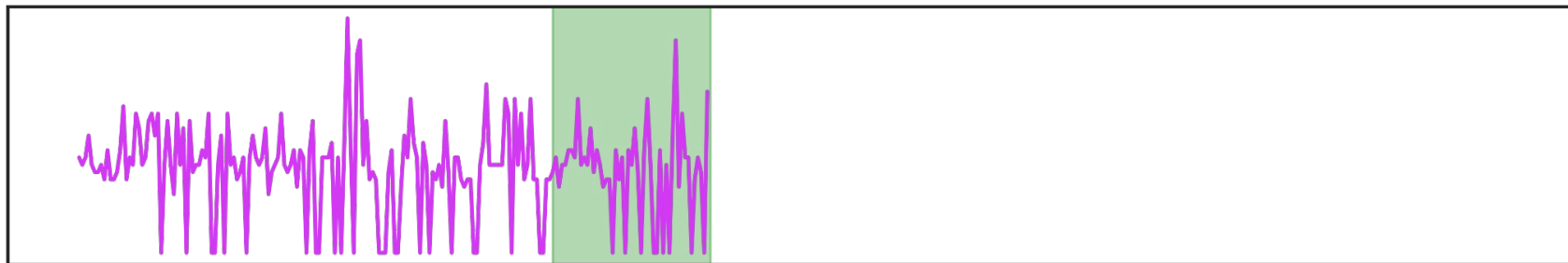
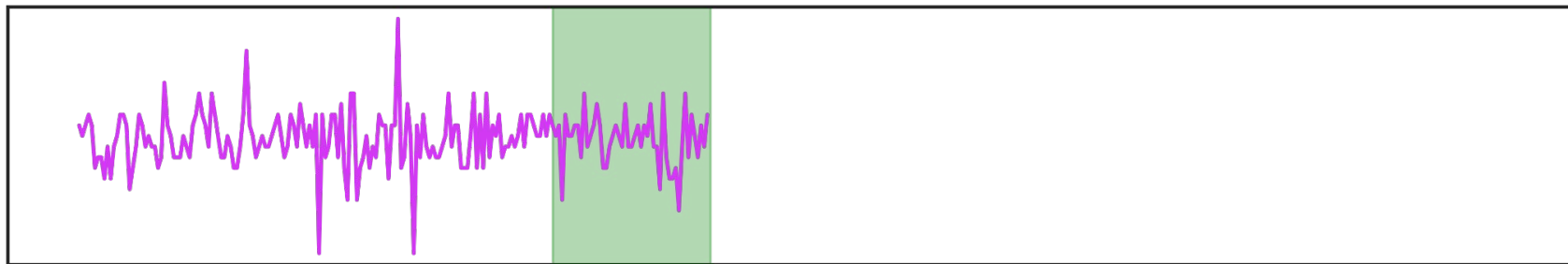
0

100

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Алгоритм в pysad

Emaad Manzoor, Hemank Lamba, and Leman Akoglu. Xstream: outlier detection in feature-evolving data streams. In *Proceedings of the 24th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining, 1963–1972*. 2018.

Резюме

Вопросы?

